GCC Bank - Command Line Banking System

C++ Project Documentation

Name: _____ Student No: _____ Date: _____
GitHub link: _____

1. Introduction

This program is a simple command line banking system written in C++. After logging in with an account number and PIN, the user can deposit money, withdraw money, send cash to another account and check their balance. The menu repeats until the user chooses Exit.

2. Features

Feature	Description
Menu and loop	The menu is shown in a do-while loop until the user chooses 5 (Exit).
Deposit	Adds money to the account. The amount must be more than 0.
Withdraw	Removes money from the account. The user cannot withdraw more than the balance.
Send Cash	Transfers money to another account. Checks that the account exists and that the balance is enough.
Check Balance	Displays the current balance.
User details	3 users (name, account number, PIN, balance) are stored in arrays.
Validation	Clear messages are shown for a wrong PIN, wrong account number, invalid menu choice or invalid amount.

3. Functions Used

Function	Parameters	Returns	Purpose
showMenu()	none	void	Prints the bank menu.
searchAccount()	int number	int	Looks for an account number and returns its position, or -1 if not found.
pinIsCorrect()	int user, int pin	bool	Checks if the PIN matches the user's PIN.
deposit()	int user	void	Adds money to the user's balance.
withdraw()	int user	void	Takes money out of the user's balance.
sendCash()	int user	void	Sends money from the user to another account.
checkBalance()	int user	void	Shows the user's balance.

4. Test Accounts

Name	Account Number	PIN	Starting Balance
Anele	2001	1234	R300.00
Mpho	2002	4321	R800.00
Zanele	2003	5555	R1200.00

5. How to Compile and Run

```
g++ bank_v2.cpp -o bank
./bank
```

6. Sample Output

```
Welcome to GCC Bank
Account number: 2001
PIN: 1234
Hello Anele!

===== GCC BANK =====
1. Deposit
2. Withdraw
3. Send Cash
4. Check Balance
5. Exit
Enter choice: 1
How much do you want to deposit? R200
Done! Money added.
Enter choice: 4
Balance: R500
Enter choice: 3
Account number to send to: 2002
How much do you want to send? R100
Sent R100 to Mpho
Enter choice: 2
How much do you want to withdraw? R5000
You don't have enough money.
Enter choice: 9
Wrong choice, pick a number from 1 to 5.
Enter choice: 5
Goodbye!
```

7. Source Code (bank_v2.cpp)

```
#include <iostream>
#include <string>
using namespace std;

// 3 users - the position in each array belongs to the same person
string userName[3] = {"Anele", "Mpho", "Zanele"};
int accountNo[3] = {2001, 2002, 2003};
int userPin[3] = {1234, 4321, 5555};
double money[3] = {300, 800, 1200};

// show the menu
void showMenu() {
    cout << endl;
    cout << "===== GCC BANK =====" << endl;
    cout << "1. Deposit" << endl;
    cout << "2. Withdraw" << endl;
    cout << "3. Send Cash" << endl;
    cout << "4. Check Balance" << endl;
    cout << "5. Exit" << endl;
    cout << "Enter choice: ";
}

// look for the account number, give back its position (or -1 if not found)
int searchAccount(int number) {
    for (int i = 0; i < 3; i++) {
        if (accountNo[i] == number) {
            return i;
        }
    }
    return -1;
}

// check if the pin is correct for this user
bool pinIsCorrect(int user, int pin) {
    if (userPin[user] == pin) {
        return true;
    }
    return false;
}

// add money
void deposit(int user) {
    double amount;
    cout << "How much do you want to deposit? R";
    cin >> amount;

    if (amount > 0) {
        money[user] = money[user] + amount;
        cout << "Done! Money added." << endl;
    } else {
        cout << "Amount must be more than 0!" << endl;
    }
}

// take money out
void withdraw(int user) {
    double amount;
    cout << "How much do you want to withdraw? R";
    cin >> amount;

    if (amount <= 0) {
        cout << "Amount must be more than 0!" << endl;
    } else if (amount > money[user]) {
        cout << "You don't have enough money." << endl;
    } else {
        money[user] = money[user] - amount;
        cout << "Done! Please take your cash." << endl;
    }
}

// send money to someone else
```

```
void sendCash(int user) {
    int number;
    double amount;

    cout << "Account number to send to: ";
    cin >> number;

    int other = searchAccount(number);

    if (other == -1) {
        cout << "That account does not exist." << endl;
    } else if (other == user) {
        cout << "You can't send money to your own account." << endl;
    } else {
        cout << "How much do you want to send? R";
        cin >> amount;

        if (amount <= 0) {
            cout << "Amount must be more than 0!" << endl;
        } else if (amount > money[user]) {
            cout << "You don't have enough money." << endl;
        } else {
            money[user] = money[user] - amount;
            money[other] = money[other] + amount;
            cout << "Sent R" << amount << " to " << userName[other] << endl;
        }
    }
}

// show balance
void checkBalance(int user) {
    cout << "Balance: R" << money[user] << endl;
}

int main() {
    int number, pin;
    int user = -1;

    cout << "Welcome to GCC Bank" << endl;

    // keep asking until the user logs in correctly
    while (user == -1) {
        cout << "Account number: ";
        cin >> number;
        cout << "PIN: ";
        cin >> pin;

        int found = searchAccount(number);

        if (found != -1 && pinIsCorrect(found, pin)) {
            user = found;
            cout << "Hello " << userName[user] << "!" << endl;
        } else {
            cout << "Wrong account number or PIN, try again." << endl;
        }
    }

    int choice;

    do {
        showMenu();
        cin >> choice;

        if (choice == 1) {
            deposit(user);
        } else if (choice == 2) {
            withdraw(user);
        } else if (choice == 3) {
            sendCash(user);
        } else if (choice == 4) {
            checkBalance(user);
        } else if (choice == 5) {
            cout << "Goodbye!" << endl;
        } else {
```

```
            cout << "Wrong choice, pick a number from 1 to 5." << endl;
        }

    } while (choice != 5);

    return 0;
}
```